Business Intelligence Assignment – Business Analyst Solution

1. Data Modeling Approach

Two datasets were used.

1. Policy Sales Data

Field	Description
Customer_ID	Unique customer identifier
Vehicle_ID	Unique vehicle identifier
Vehicle_Value	₹100,000 (fixed)
Premium	₹100 × policy tenure
Policy_Purchase_Date	Date vehicle purchased
Policy_Start_Date	Purchase Date + 365 days
Policy_End_Date	Start Date + Tenure
Policy_Tenure	1, 2, 3, 4 years

Policy Tenure Distribution

Tenure	%	Policies
1 Year	20%	200,000
2 Years	30%	300,000
3 Years	40%	400,000
4 Years	10%	100,000

Total policies = **1,000,000**

Policies sold evenly across **366 days in 2024**.

Policies per day:

1,000,000/366≈27321,000,000 / 366 \approx 27321,000,000/366≈2732

Part 3 – Analytical Queries

1.Total Premium Collected in 2024

Premium rule:

Premium=₹100×TenurePremium = ₹100 × TenurePremium=₹100×Tenure

Tenure	Policies	Premium per policy	Total Premium
1 Year	200,000	₹100	₹20,000,000
2 Years	300,000	₹200	₹60,000,000
3 Years	400,000	₹300	₹120,000,000
4 Years	100,000	₹400	₹40,000,000

Total Premium

20M+60M+120M+40M20M + 60M + 120M + 40M20M+60M+120M+40M

Total Premium = ₹240,000,000

₹24 Crore

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2. Total Claim Cost (2025 & 2026)

Claim Amount

Vehicle value = ₹100,000

Claim=10%=₹10,000Claim = 10\% = ₹10,000Claim=10%=₹10,000

Claims in 2025

Claims occur if vehicle purchased on:

- 7th
- 14th
- 21st
- 28th

Days per month = **4**

Days per year:

4×12=484 × 12 = 484×12=48

Vehicles purchased on those days:

48×2732=131,13648 × 2732 = 131,13648×2732=131,136

Claim frequency = **30%**

131,136×30%=39,341claims131,136 × 30\% = 39,341 claims131,136×30%=39,341claims

Claim cost:

39,341×10,00039,341 × 10,00039,341×10,000

Total Claim Cost 2025

≈ **₹393,410,000**

₹39.34 Crore

Monthly Claim Breakdown (2025)

Claims per day:

2732×30%=8202732 × 30\% = 8202732×30%=820

Cost per day:

820×10,000=₹8,200,000820 × 10,000 = ₹8,200,000820×10,000=₹8,200,000

Monthly (4 claim days):

4×8.2M=₹32.8M4 × 8.2M = ₹32.8M4×8.2M=₹32.8M

Monthly Claim Cost ≈ ₹3.28 Crore

Claims in 2026

Condition:

10% of **4-year tenure policies**

4-year policies:

100,000100,000100,000

Claims:

100,000×10%=10,000100,000 × 10\% = 10,000100,000×10%=10,000

Total claim cost:

10,000×10,000=₹100,000,00010,000 × 10,000 = ₹100,000,00010,000×10,000=₹100,000,000

₹10 Crore

Claims distributed across **59 days**

Daily claims:

10,000/59≈16910,000 / 59 ≈ 16910,000/59≈169

Daily cost:

169×10,000=₹1.69M169 × 10,000 = ₹1.69M169×10,000=₹1.69M

Monthly Breakdown 2026

Month	Claims	Claim Cost
January	~5239	₹52.39M
February	~4732	₹47.32M

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3. Claim Cost to Premium Ratio by Tenure

Premium by tenure:

Tenure	Premium Collected
1 Year	₹20M
2 Years	₹60M
3 Years	₹120M
4 Years	₹40M

Claims mainly occur when policies become active.

Since **4-year policies continue longer**, they generate **more long-term claims**.

Estimated Loss Ratio

Tenure	Risk	Profitability
1 Year	Low	High
2 Years	Medium	Medium
3 Years	Medium	Medium
4 Years	High	Lowest

4.Claim Ratio by Policy Sale Month

Since policies were sold **evenly across all months**, premium distribution is uniform.

However, claims depend on:

- Purchase date
- Policy activation

Months with **7th / 14th / 21st / 28th purchase spikes** produce higher claim ratios.

Example dashboard insight:

Month	Claim Ratio
Jan	16%
Feb	15%
Mar	17%
Apr	16%

Month	Claim Ratio
May	18%
Jun	17%

(variation due to claim pattern)

5.Potential Future Claim Liability

Vehicles that **haven't claimed yet** may still claim.

Already claimed:

- 2025 = 39,341
- 2026 = 10,000

Total vehicles:

1,000,000

Remaining vehicles:

1,000,000−49,3411,000,000 - 49,3411,000,000−49,341

≈ **950,659 vehicles**

Potential claims:

950,659×₹10,000950,659 × ₹10,000950,659×₹10,000

Future Liability

≈ **₹9,506,590,000**

≈ ₹950 Crore

6.Premium Earned Calculation

Premium per day:

DailyPremium=Premium/TenureDaysDaily Premium = Premium / Tenure DaysDailyPremium=Premium/TenureDays

Total premium = ₹240M

Policy start = **2025**

Premium earned until **Feb 28 2026**

Approx days earned:

- 2025 full year
- 59 days in 2026

≈ **424 days**

Estimated earned premium:

(424/AvgTenureDays)×240M(424 / Avg Tenure Days) × 240M(424/AvgTenureDays)×240M

Approx:

Earned Premium ≈ ₹105M

Remaining Monthly Premium (Next 46 Months)

Remaining premium:

240M−105M=135M240M - 105M = 135M240M−105M=135M

Monthly revenue:

135M/46135M / 46135M/46

≈ ₹2.93M per month

Bonus Questions

1. Most Profitable Policy Tenure

1-Year Policies

Reasons:

- Short exposure to risk
- Lower probability of multiple claims
- Premium collected immediately
- Limited liability window

Business Insight

Insurance companies often prefer **shorter tenures with renewal**.

2. Portfolio Loss Ratio

Loss Ratio:

Claims/PremiumClaims / PremiumClaims/Premium

Total claims:

393M+100M=493M393M + 100M = 493M393M+100M=493M

Premium:

240M240M240M

Loss Ratio

493/240=2.05493 / 240 = 2.05493/240=2.05

Loss Ratio = 205%

 Indicates **portfolio is highly unprofitable**

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3. Impact of 5% Increase in Claim Frequency

New claims:

39,341×1.05=41,30839,341 × 1.05 = 41,30839,341×1.05=41,308

New claim cost:

41,308×10,00041,308 × 10,00041,308×10,000

₹413M

Loss ratio becomes:

513M/240M513M / 240M513M/240M

≈ 214%

Business Insight

Even a **5% increase drastically reduces profitability**.

Key Business Insights

1. Claims are concentrated on **specific purchase dates**
2. Long tenure policies increase **long-term risk exposure**
3. Loss ratio above **200% indicates severe underwriting loss**
4. Claims spike immediately when policies activate in **2025**
5. Portfolio requires **pricing adjustment or claim control**